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Q1.(Creating Tuples)

```
numbers = (10, 20, 30, 40, 50)
```

```
mixed = (100, "Python", 3.14, True)
```

```
empty1 = ()
```

```
empty2 = tuple()
```

```
single = (99,)
```

```
print("Numbers Tuple:", numbers)
```

```
print("Mixed Tuple:", mixed)
```

```
print("Empty Tuple 1:", empty1)
```

```
print("Empty Tuple 2:", empty2)
```

```
print("Single Element Tuple:", single)
```

```
print("\nTypes:")
```

```
print("numbers:", type(numbers))
```

```
print("mixed:", type(mixed))
```

```
print("empty1:", type(empty1))
```

```
print("empty2:", type(empty2))
```

```
print("single:", type(single))
```

Q2.(Tuple Packing)

```
name = "Vaishnavi"
```

```
age = 20
```

```
city = "Pune"
```

```
student = name, age, city
```

```
print("Tuple:", student)
```

```
print("Type:", type(student))
```

```
student_name, student_age, student_city = student
```

```
print("\nAfter Unpacking:")
```

```
print("Name:", student_name)
```

```
print("Age:", student_age)
```

```
print("City:", student_city)
```

Q3.(Indexing and Negative Indexing)

```
colors = ('red', 'green', 'blue', 'yellow', 'purple', 'orange')
```

```
print("First element:", colors[0])
```

```
print("Third element:", colors[2])
```

```
print("Last element:", colors[-1])
```

```
print("Second last element:", colors[-2])
```

```
index = int(input("Enter an index number: "))
```

```
print("Element at index", index, "is:", colors[index])
```

Q4.(Slicing Tuples)

```
numbers = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
```

a) Elements from index 2 to 7

```
print("a) Index 2 to 7:", numbers[2:8])
```

b) First 5 elements

```
print("b) First 5 elements:", numbers[:5])
```

c) Last 4 elements

```
print("c) Last 4 elements:", numbers[-4:])
```

d) Every second element

```
print("d) Every second element:", numbers[::2])
```

e) Entire tuple in reverse order

```
print("e) Reverse tuple:", numbers[::-1])
```

Q5.(Nested Tuples)

```
matrix = (
```

```
    (1, 2, 3),
```

```
    (4, 5, 6),
```

```
    (7, 8, 9)
```

```
)
```

```
print("First row:", matrix[0])
```

```
print("Second row, Third column:", matrix[1][2])
```

```
print("Third row, First column:", matrix[2][0])
```

Q6.(Iterating and Unpacking)

```
student = ('Rahul', 20, 'Computer Science', 'A')
```

```
print("Tuple Elements:")  
  
for item in student:  
    print(item)  
  
name, age, branch, grade = student  
  
print("\nStudent Details:")  
  
print("Name:", name)  
  
print("Age:", age)  
  
print("Branch:", branch)  
  
print("Grade:", grade)
```

Q7.(count() Method)

```
grades = ('A', 'B', 'A', 'C', 'A', 'B', 'D', 'A', 'B')  
  
print("Count of A:", grades.count('A'))  
  
print("Count of B:", grades.count('B'))  
  
grade = input("Enter a grade (A/B/C/D): ")  
  
print("Count of", grade, "is:", grades.count(grade))
```

Q8.(index() Method)

```
fruits = ('apple', 'banana', 'cherry', 'banana', 'mango', 'apple')  
  
print("First index of banana:", fruits.index('banana'))  
  
print("First index of banana from index 2:", fruits.index('banana', 2)) try:  
  
    index = fruits.index('kiwi')
```

```
    print("Index of kiwi:", index)
except ValueError:
    print("Not found")
```

Q9.(Immutability of Tuples)

```
coordinates = (10, 20)
print("Original Tuple:", coordinates)
coordinate_list = list(coordinates)
coordinate_list[0] = 100
coordinate_list.append(30)
coordinates = tuple(coordinate_list)
print("Modified Tuple:", coordinates)
```

Q10.(Mini Project - Combined Concepts)

```
name = input("Enter Student Name: ")
roll_no = int(input("Enter Roll Number: "))
mark1 = int(input("Enter Subject 1 Marks: "))
mark2 = int(input("Enter Subject 2 Marks: "))
mark3 = int(input("Enter Subject 3 Marks: "))
student1 = (name, roll_no, mark1, mark2, mark3)
print("\n----- Student Record -----")
print(student1)
name, roll_no, mark1, mark2, mark3 = student1
```

```
print("\n----- Student Details -----")

print("Name      :", name)

print("Roll No   :", roll_no)

print("Subject 1  :", mark1)

print("Subject 2  :", mark2)

print("Subject 3  :", mark3)

search_mark = int(input("\nEnter a mark to count: "))

print("Count of", search_mark, ":", student1.count(search_mark))

student1= ("Rahul", 102, 75, 80, 85)

students = (student1, student2)

print("\n----- All Student Records -----")

print(students)

print("\nFirst Student Name :", students[0][0])

print("First Student Roll No :", students[0][1])

print("Second Student Name :", students[1][0])

print("Second Student Subject 2 Marks :", students[1][
```